Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **AS FinTech Developer**
b. Team leader name: **Abhinav Mukwane & Sourabh Thakre**
c. Problem Statement: **Intelligent Risk, Fraud & Financial Experiences (BFSI).**
FinRiskLens AI — fraud detection and risk analysis for every MSME customer, from shared data to a grounded AI credit decision

Brief about the idea

₹36,014 cr

involved in bank frauds in FY25

₹33,148 cr

of it in loans — the largest category

Source: RBI Annual Report 2024-25 (date of reporting)

MSME loans are decided on shared data and uploaded documents. A statement or ITR can be edited in minutes; one fabricator can supply many applicants.

FinRiskLens gives the bank one fraud and risk page per customer. Customers share bank data through Account Aggregator, fetch GST, ITR and MCA, and upload documents. Gemini reads every page, code checks the arithmetic and the bank's own record, and the network lens links related customers.

Every finding links to its evidence. A Clear customer gets **a grounded AI credit decision, sent to the LOS, ULI (OCEN) or ONDC.**

Every source, one page

Udyam, GST, bank, ITR, MCA and uploads, section by section

The bank's own record

Uploaded statements against the Account Aggregator feed

The network

Shared directors, contacts and files; circular GST trading

Grounded AI decisions

Investigation and credit decision; every figure traced

Opportunities

How it differs from existing tools

- **Statement analysers** look at one document — FinRiskLens combines every source of a customer and returns findings with evidence
- **Arithmetic checks** pass a careful forgery — FinRiskLens also compares it with the bank's own record
- **Single-customer review** misses organised fraud — the network lens links related customers

How it solves the problem

- **Fast:** a customer is reassessed as soon as a source arrives
- **Verifiable:** every finding opens the document at its page and row
- **Decisive:** the officer confirms or dismisses each finding; a Clear customer goes straight to a loan case
- **Policy-aware:** findings and credit decisions cite the lender's credit policy

USP

- **Grounded AI:** a claim or figure that cannot be traced is set aside — and shown
- **Bank-record cross-check** through Account Aggregator
- **Linked customers** through directors, contacts, files and GST invoices
- **Risk to loan:** AI credit decision, then LOS, ULI (OCEN) or ONDC
- **Documents stay in India:** read and stored in Mumbai

Who gains

Credit officers	Fraud teams	Honest MSMEs
One page per customer, findings with evidence	Linked customers across the portfolio	A Clear verdict — a faster loan

List of features offered by the solution

Customer data, one place

- Business Registration (Udyam); Banking Data by Account Aggregator consent
- Tax & Sales (GST), Income Tax (ITR), Corporate Registry (MCA); uploads
- Reassessed whenever a source arrives

Document reading

- Gemini Flash in Mumbai reads every page in parallel
- Numbers copied as printed, never corrected
- A page that does not add up is read again

Checks on every source

- Nine checks: balances, totals, slips, file history
- Statement vs the bank's Account Aggregator record
- Bank risk flags: pass-through, cash, bounces

Linked customers

- Same file, template or salary account
- Same director, mobile, email or address
- Circular trading in GST invoices

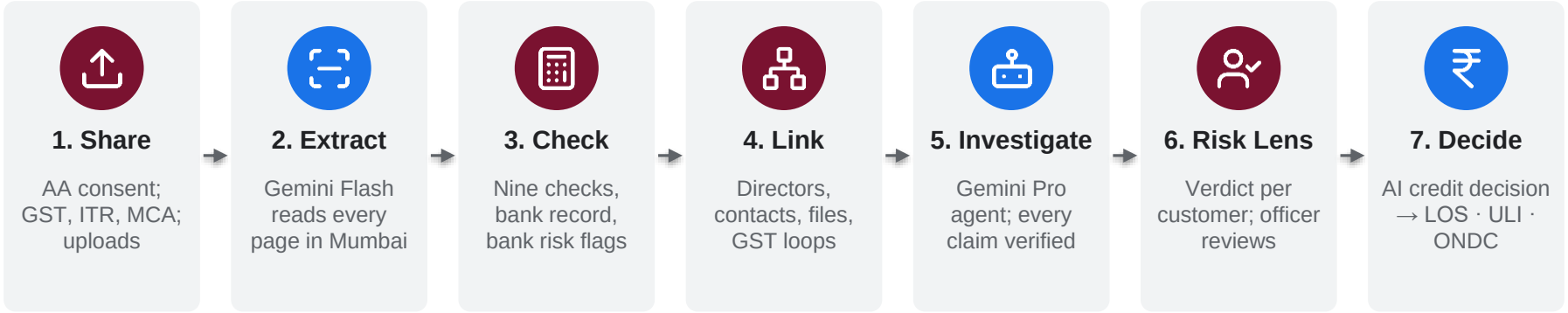
AI investigator, grounded

- Gemini Pro with tools bound to one customer
- Every claim cites rows; amounts must trace
- Rejected claims shown, never hidden

AI credit decision

- Credit capacity (FOIR) computed in code
- Gemini Pro decides within it, citing policy
- Accepted decisions go to LOS, ULI or ONDC

Process flow diagram



Customer (MSME)

Registers with the Udyam number, shares bank data through Account Aggregator, fetches GST, ITR and MCA, uploads documents

Credit officer

One Risk Lens per customer: reviews each finding, generates and accepts the credit decision, sends the loan case

LOS · ULI · ONDC

Receive the accepted loan case with its risk assessment and the AI credit decision

Wireframe — a finding on the Risk Lens

HIGH Tampering STATEMENT_VS_AA_MISMATCH Deterministic

Statement disagrees with the bank's own record

91 of 93 rows agree with the bank's Account Aggregator record, with 2 amounts changed and 4 transactions left out. Net effect: ₹3,65,800 more in credits and ₹1,30,000 less in debits than the bank recorded.

↳ **Assess the customer on the Account Aggregator data and ask them to explain every difference.**

↗ statement.pdf, p. 2 · row 28 — bank recorded ₹1,09,500

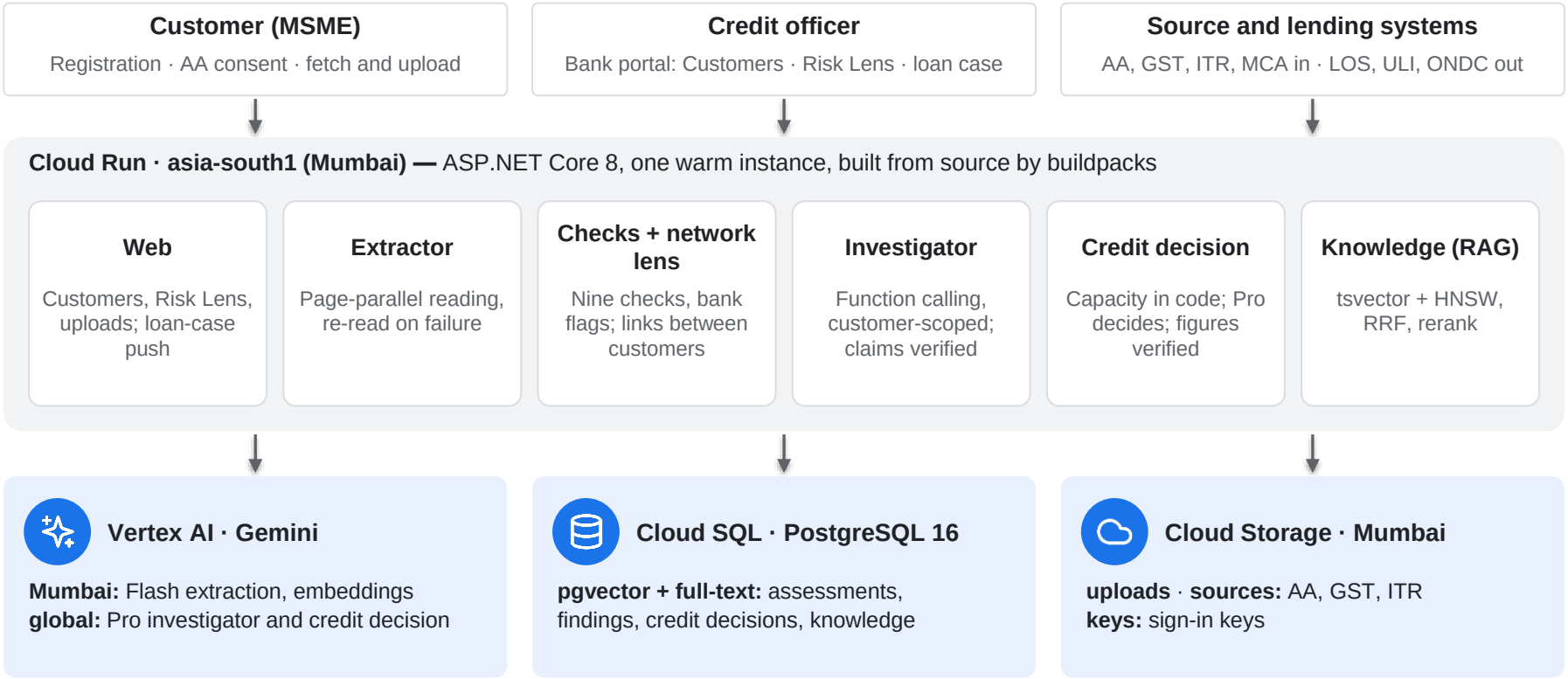
↗ statement.pdf, p. 3 · row 54 — bank recorded ₹73,400

↗ statement.pdf, p. 1 · 05 Jun NACH EMI ₹32,500 — not on the statement

Note for the review (optional) | Confirm | Dismiss

1. **Severity, category, type and source**
   Deterministic or AI · verified
2. **What is wrong, in rupees**
   Plain language a credit officer can act on
3. **What to do next**
   Names the policy clause when one applies
4. **Evidence links**
   Open the document at that page; rows, fields or policy
5. **Review in one click**
   A confirmed finding blocks the loan case; a dismissed one does not

Architecture diagram of the proposed solution



Technologies to be used in the solution

Google Cloud

- Vertex AI — Gemini 3.5 Flash (extraction), Gemini 3.1 Pro (investigator, credit decision), Gemini 3.8 Flash (rerank), gemini-embedding-001
- Cloud Run (buildpacks, no Dockerfile) · Cloud Build
- Cloud SQL for PostgreSQL 16 + pgvector · Cloud Storage

AI techniques

- Multimodal structured output: PDF and image to a fixed schema
- Function-calling agent with tools bound server-side to one customer
- Grounding verifier · hybrid RAG: full-text + HNSW vectors, reciprocal rank fusion, LLM rerank

Application

- .NET 8 · ASP.NET Core MVC · Razor + Bootstrap 5
- EF Core 8 + Npgsql · Autofac · Serilog
- PdfPig (PDF metadata) · background worker · LOS, ULI (OCEN) and ONDC (beckn) payloads

Data, quality and trust

- Account Aggregator on the Finvu sandbox; Udyam, GST, ITR and MCA as sample data in the demo
- QuestPDF synthetic documents with ground truth · 70 xUnit tests
- Secrets outside the public repo · encrypted DB connections only

Estimated implementation cost (optional)

₹ 18 L
One-time — production hardening, security and compliance audit

₹ 32 L / yr
Recurring — Google Cloud, Gemini usage, data access, support

~ ₹ 1.14 Cr
3-year total cost of ownership

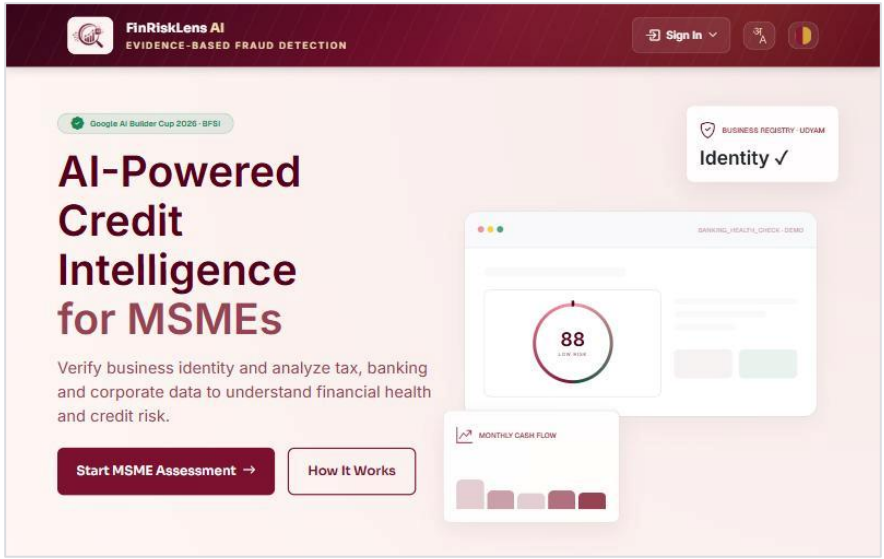
Estimate for one lender, about 1 lakh customers assessed a year, on Google Cloud in Mumbai

Cost head	Type	Estimate (INR)
Production hardening — 3-member team, about 4 months	One-time	₹ 14 L
Security testing and RBI / DPDP compliance audit	One-time	₹ 4 L
Google Cloud — Cloud Run (HA), Cloud SQL (HA), Cloud Storage	Annual	₹ 8 L
Vertex AI — Gemini extraction, investigation, credit decisions, embeddings	Annual	₹ 6 L
Data access — Account Aggregator and registry APIs	Annual	₹ 10 L
Maintenance, support and model re-evaluation	Annual	₹ 8 L

Snapshots of the prototype — the customer

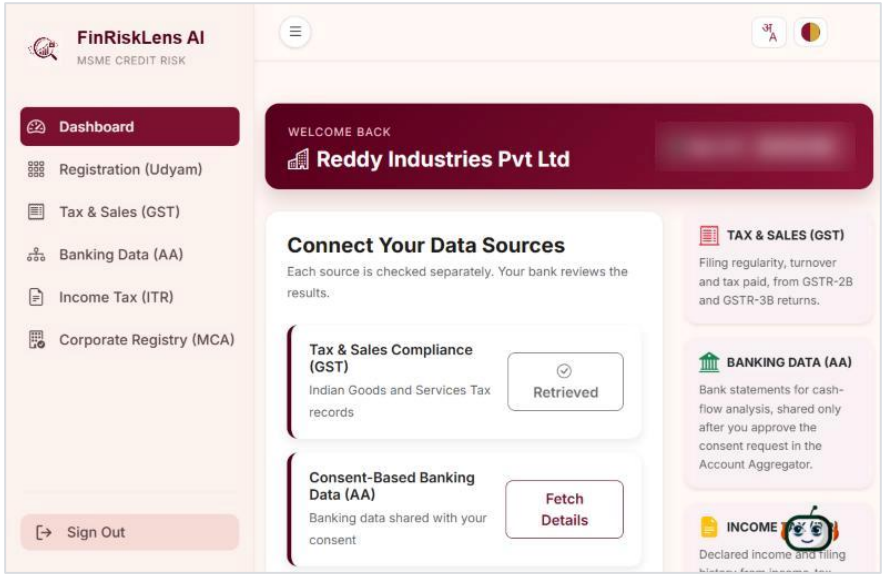
Home page

Customers start here with their Udyam number



Customer dashboard

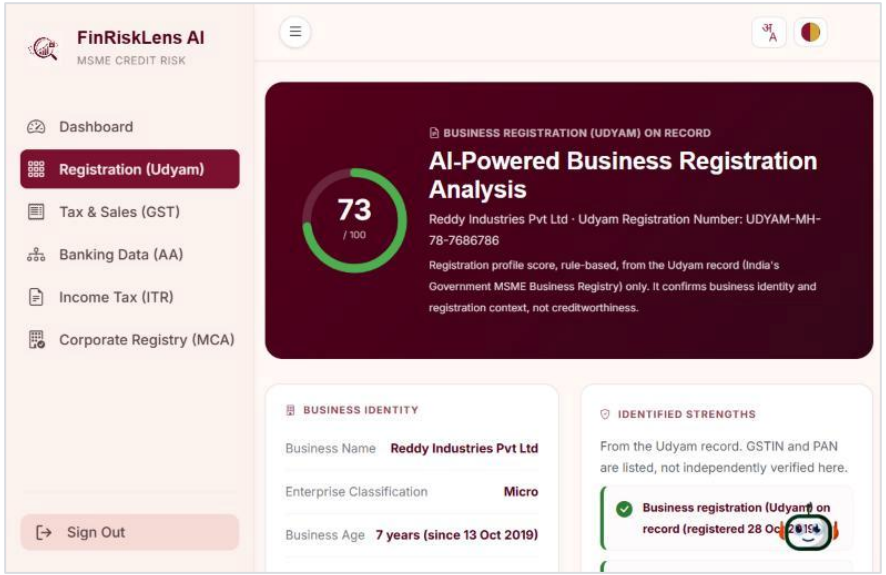
After sign-in: connect each data source, upload documents



Snapshots of the prototype — each source analysed

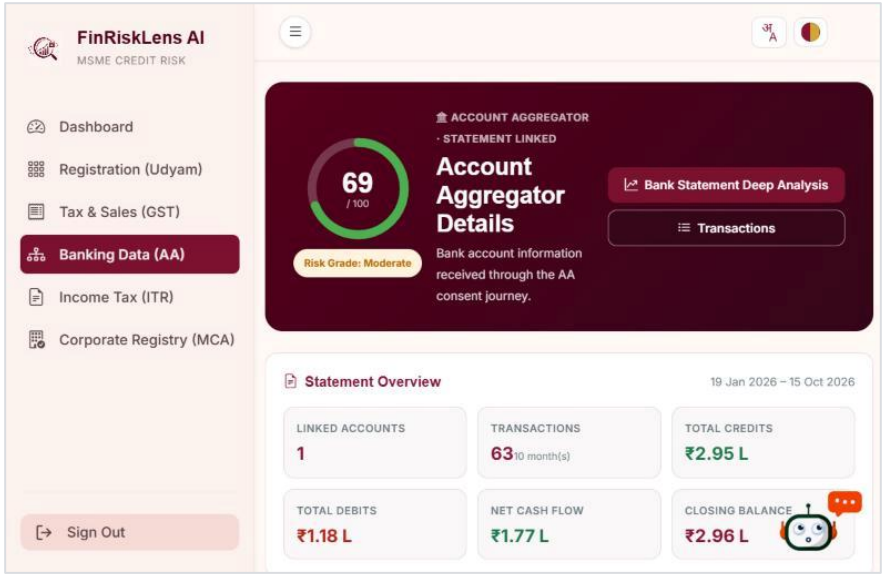
Business Registration (Udyam)

The registration profile, from the Udyam record



Banking Data (AA)

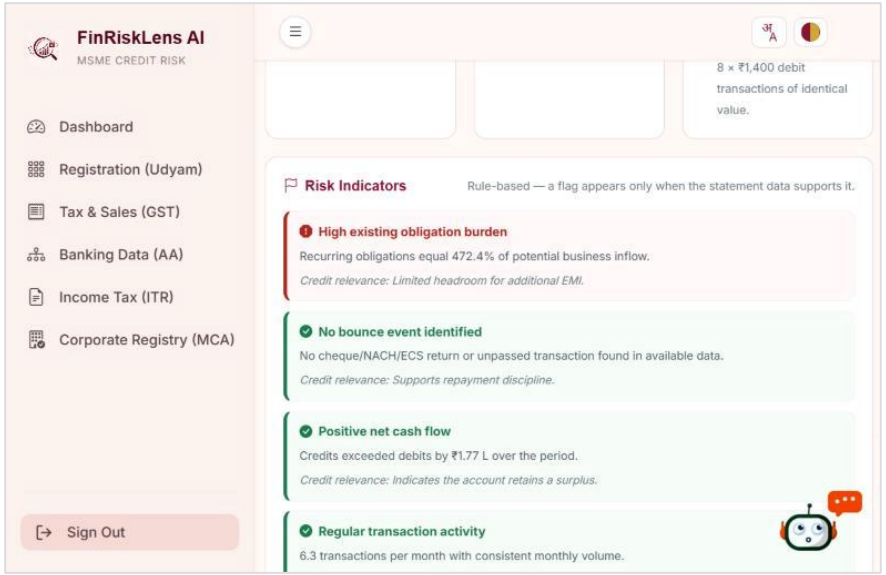
The bank's own record, shared by consent



Snapshots of the prototype — each source analysed

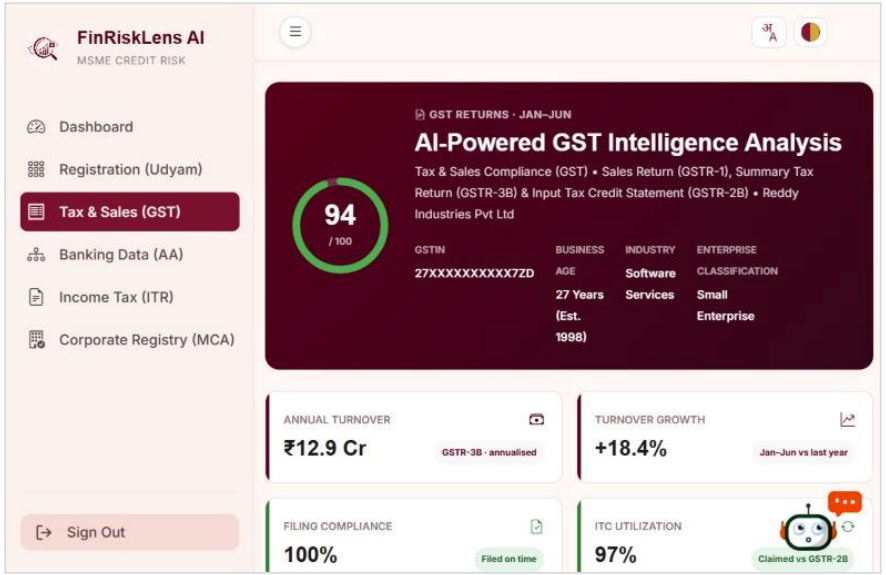
Bank statement risk indicators

A flag only where the statement data supports it



Tax & Sales Compliance (GST)

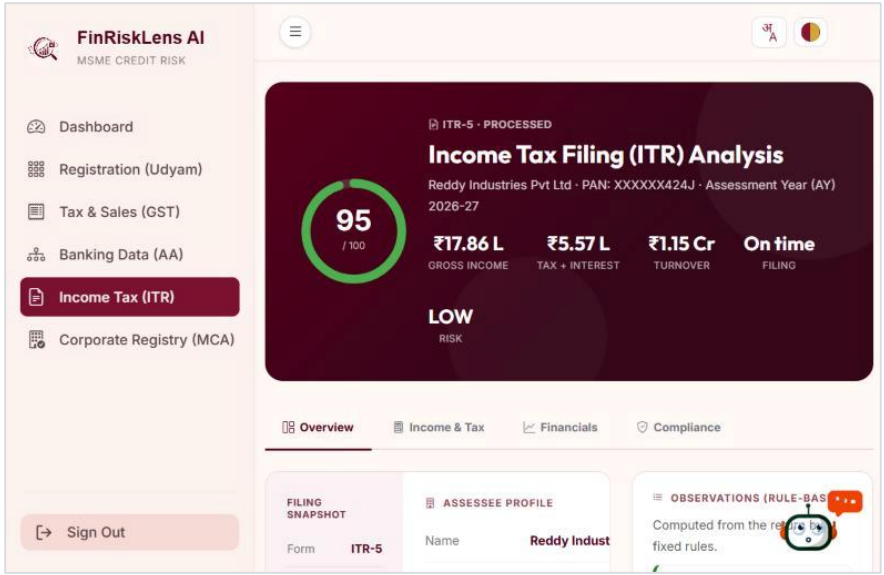
Filing and turnover analysis — sample data in the demo



Snapshots of the prototype — each source analysed

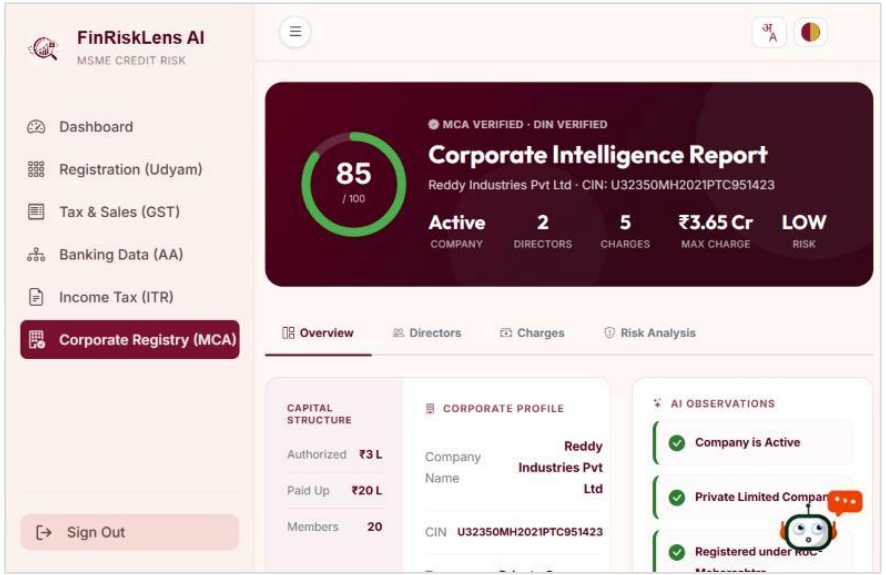
Income Tax Filing (ITR)

Income, tax and filing — sample data in the demo



Corporate Registry (MCA)

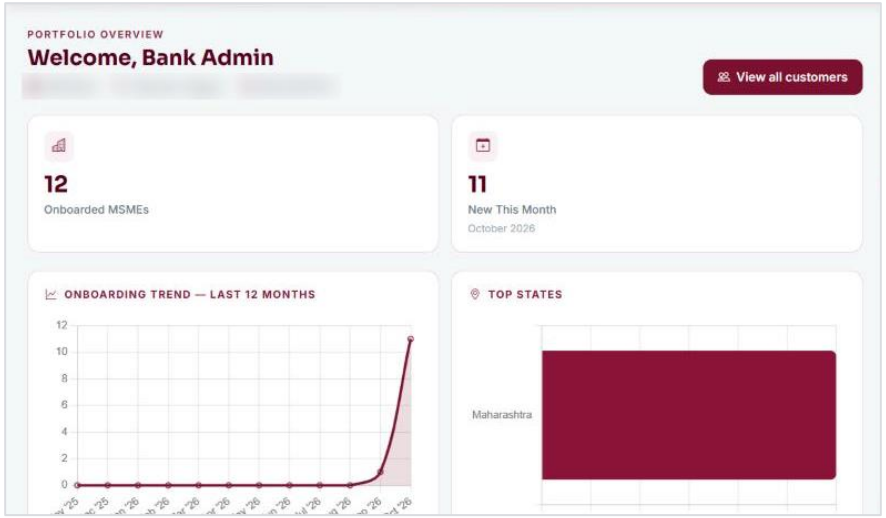
Company, directors and charges — sample data in the demo



Snapshots of the prototype — the bank

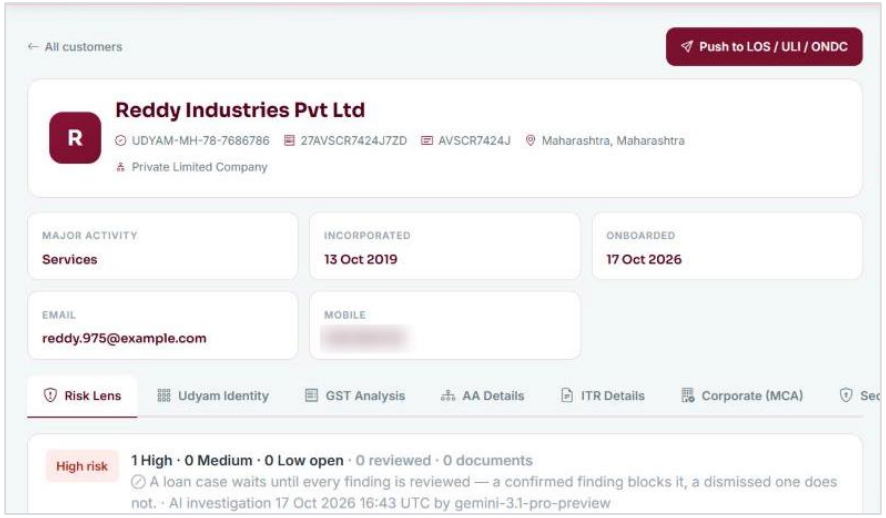
Bank dashboard

The onboarded portfolio at a glance

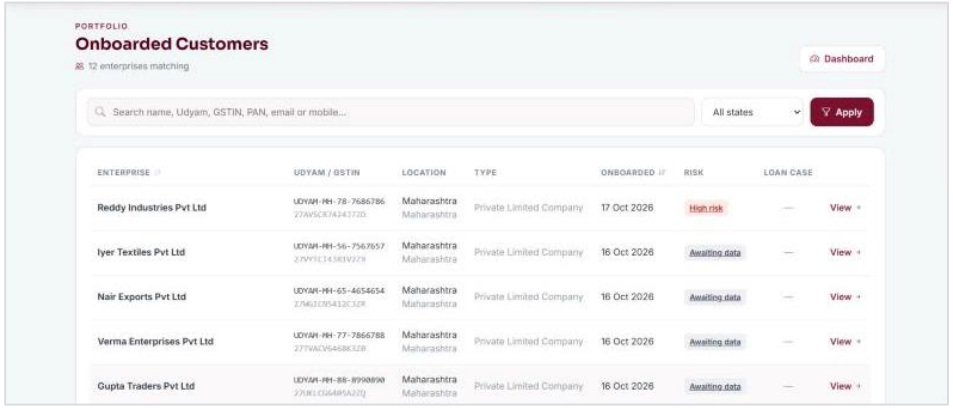


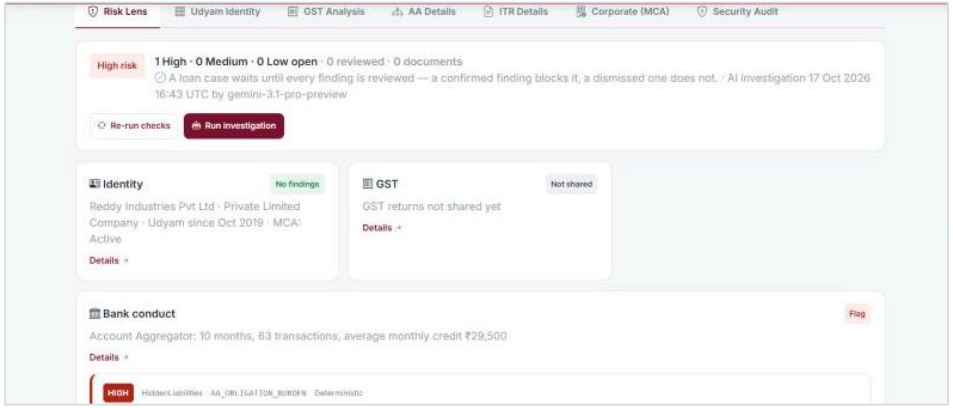
One page per customer

Opens on the Risk Lens; every source in its own tab



Snapshots of the prototype — fraud detection per customer





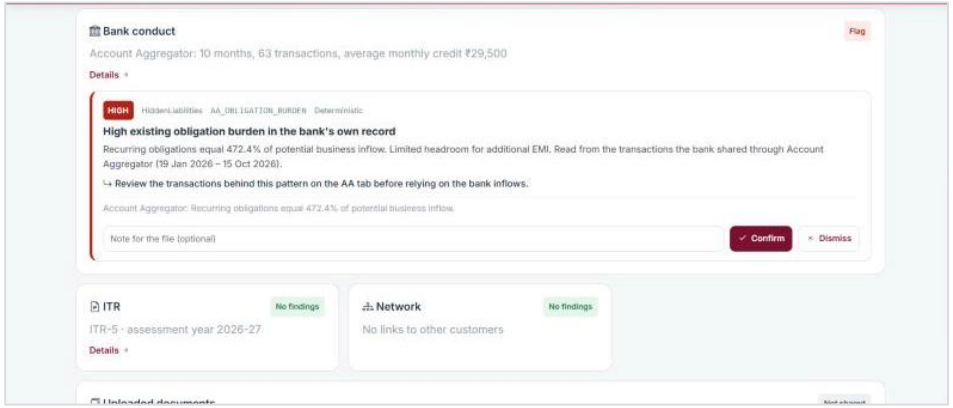
Customers

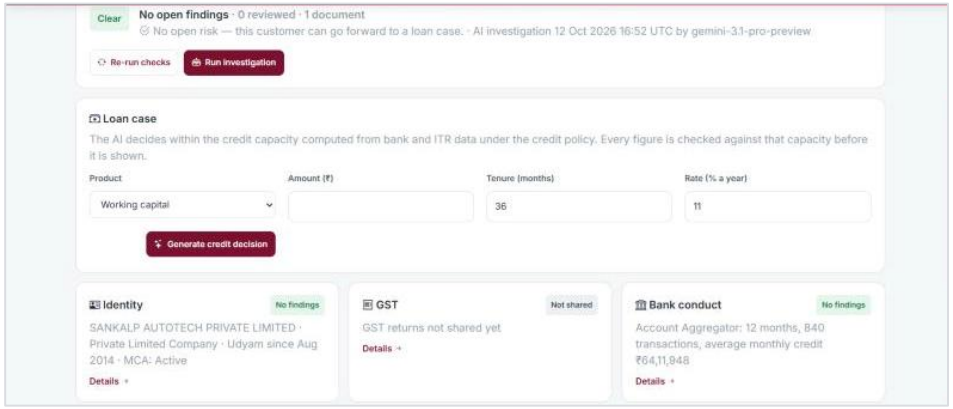
A risk status for every customer — Reddy Industries reads High risk

Risk Lens

The verdict, then one section per source — Bank conduct is flagged

Snapshots of the prototype — fraud detection per customer





A finding with its evidence

Read from the bank's own Account Aggregator record; the officer confirms or dismisses it

Clear → loan case

A customer with nothing open goes on to the AI credit decision and the LOS

Prototype Performance report/Benchmarking

99.8–100%
of 1,298 statement rows read exactly — two live runs

20 / 20
demo assessments raise exactly their expected findings

0
false alarms on genuine applicants — checks and AI

5 / 5
expected AI findings found by the investigator

How it is measured: synthetic documents with known ground truth, read by live Gemini, scored row by row, checks run on Gemini's own reading

Check	Manual review	FinRiskLens
Statement arithmetic	Spot-checked by eye	Every row and page: balance, carry-over, totals
Statement vs bank	Ask for a bank-certified copy	Automatic when Account Aggregator is linked
Edited files	Rarely examined	Producer, creator and edit dates read on upload
Links across customers	One customer at a time	Shared file, account, director or contacts; GST loops
Credit decision	FOIR worked out by hand	Capacity computed in code; every AI figure traced
Speed	Analyst time per file	10–60 s per statement · 12–43 s per investigation

Additional Details/Future Development (if any)

Registry cross-checks

- GST, ITR and bank turnover reconciled
- Name, PAN and address across every source
- Password-protected and scanned statements

Evidence highlighting

- Highlight the exact row on the PDF page
- Bounding boxes already in the extraction contract
- Side-by-side PDF and bank record view

Case memory and policy Q&A

- Confirmed findings feed the knowledge base
- Policy questions answered with citations
- Refusal when the policy is silent

Scale and integration

- Entity link tables for portfolio-scale customer links
- Live GST, ITR, MCA and OFAC/AML screening via licensed providers
- Per-lender policies and thresholds

Provide links to your:

- GitHub Public Repository: https://github.com/abhinavmukwane/FinRiskLensAI-RAG-V2
- Demo Video Link (3 Minutes): **add the link after uploading the video**
- Final Product Link: https://finrisklensai.cloud.run

For the evaluation panel

Bank portal: https://finrisklensai.cloud.run/Auth/BankLogin — credentials are shared with the panel directly.

All demo data is synthetic. GST, ITR and MCA are generated samples; OFAC/AML screening is a demo.

In the repository

README — how it works and every measurement · Doc/12_CUSTOMER_RISK_LENS.md — the customer flow · Doc/ARCHITECTURE.md

Run it yourself: dotnet test (70 tests) · DemoData eval (live extraction scored against ground truth)

Try it on the live site

As a customer (MSME)

1 Start MSME Assessment on the home page: register with the Udyam number, verify by email code

2 Connect data sources: bank data by Account Aggregator consent; GST, ITR and MCA

3 Upload bank statements or other documents

4 Each source page explains the business's own data; the assistant answers questions on it

As the bank (panel sign-in)

1 Dashboard: the onboarded portfolio

2 Customers: a risk status for each — Clear, Review, High risk or Awaiting data

3 Open a customer: the Risk Lens — verdict, a section per source, each finding with its evidence; confirm or dismiss

4 A Clear customer: the AI credit decision, then the loan case to LOS, ULI (OCEN) or ONDC

Demo data: every business is synthetic; GST, ITR and MCA are generated samples; Account Aggregator runs on the Finvu sandbox; OFAC/AML screening is a demo. Bank sign-in details are on the links slide.

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Thank you

